

# RF and Microwave Engineering (EX752 / EX716)

## Concise PYQ

**Notes:** TU IOE PYQ 2069–2082, 24 papers. Regular = **bold**, Back = normal; “Regular / Back” counts as Regular. BEX (EX752 / EG785, IV/II) = normal font, BEI (EX716, IV/I) = **mono**. Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

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## 1. Introduction

- Microwave frequency bands (IEEE) + applications; comm protocols of two / three / five systems (4)(2+4)(3+3)(2+6)(4+6)(5+5)(3+3+4)
- Conventional Vs RF/microwave circuits; lumped Vs distributed; passive components at low Vs microwave bands (4+2)(3+3)(8)(10)
- (+)(-) of microwaves Vs acoustic / seismic / conventional / lower-freq (5)(6)(2+4)(4+4)
- Why S-parameter based analysis is a must in microwave network analysis (5+3)

## 2. RF & M/W Transmission Lines

- Microwave TL Vs conventional; microstrip Vs strip-lines; TL behaviour at low Vs microwave bands (4)(5)(10)
- Admittance / immittance chart; sketch + duality parameters + compare scales (2)(10)(4)(5)
- Smith Chart on 50Ω line with 75+j100:  $\Gamma_L$ , VSWR,  $Z_{in}$ @0.375λ, shortest purely-resistive length (10)
- Single-stub matching ( $Z_L=80+j100$ ;  $\Gamma_L=0.5\angle 51^\circ$ ; 120-j160/100Ω; 78.27+j60.93/75Ω) +Smith, +S-matrix (2+8)(8)(5+3+2)(4+4+2)
- Double-stub matching (3λ/8) +Fig/+S-matrix; various loads (8)(10)(12)(8+2)(10+2+2)
  - $Z_L=300+j300$ , 80+j180@3GHz, 190+j110@10GHz, 110+j110/100Ω
  - 60-j80/50Ω, 109+j120/75Ω coax,  $\Gamma=0.64\angle 58^\circ$ /100Ω,  $\Gamma=0.45\angle 60^\circ$ @10GHz,  $\Gamma=0.33\angle 66^\circ$ /75Ω patch
  - admittance 0.00813+j0.0065 Ū/50Ω; short-ckt 75+j75, 0.4+j0.85 (3+15)(10+2)
  - (+) of double over single stub; matching techniques + S-matrix of perfect match (2+8)(4+6)

## 3. RF & M/W Network Theory

- S-parameters (I); Vs h-parameters; return & insertion loss (4)(4+4)(2)(5+5)
- S-parameters — two-port (derive); three-port network + characteristic parameters +Eg (2+6)(4+5)(4+4)(4+6)(5+2)
- S-matrix analysis: reciprocal / lossless / return loss / reflected power; S-matrix of an Amplitude Modulator (3+1+1+1+2)(3+7)
- Passive device from S-matrix (4-port); power splitter; E/H-Tee & Duplexer S-matrices (5)(6)(8)(10)
- Two radar transmitters needing double power; ASR duplexer at half input power (6)(8)(10)
- Differences between two given S-Matrix sets (4)

## 4. RF/Microwave Components & Devices

- Waveguide — rect TE/TM field equations; modes + parameters (7+3)(8+2)(8)(10)
  - TE<sub>10</sub> Vs TE<sub>20</sub> cutoff; prove TM<sub>01</sub>/TM<sub>10</sub> absent; lowest TM cutoff is TM<sub>11</sub>; TE<sub>10</sub> dominant when  $b>a$  (2)(3)
  - dominant Vs degenerate modes; dominant modes (rect & circular) + 2.254 cm WG at 6 GHz?;  $a=3b$  dominant mode; rect WG (assumed dims): cutoff,  $v_p$ ,  $\lambda_g$ ,  $Z_0$  (2+2)(1+1+2)(6)
  - circular waveguide TE/TM field; air-filled rect (cutoff/velocities/ $Z_0$ ) +directional coupler (10)(7)(6+4)
- Tee junctions — Magic Tee (S-params, +derive); E-plane Vs H-plane; hybrid tee (terminated E/H/co-planar arms) (3+3)(4)(5)(8)(4+4+2)

– two Magic Tees joined at E-arms (derive S-matrix); combined effect in H-plane; magic tee with both arms shorted (10)(5)(4)

- Directional couplers; circulators; cavity resonators; microstrips (6)(5)(4)(5)
- Probe- Vs loop-coupling; Gunn diode; MASER (5)(4)

## 5. Microwave Generators

- Transit-time effect + two-cavity klystron; bunching (density modulation); bunching + field effect (2+8)(2)(2+6)(2+2+4)
- Klystron oscillator/amplifier (multi-cavity); cross-field effect; velocity modulation + reflex klystron +Fig (2+7)(2+8)(4)(5)(10)
- Magnetron construction/operation; cross-field (45°/60°/90° phase) as power amp; circular beam-type cross-field amplifier (2+8)(10)(6+2)(8)(3+7)(7)
- TWT construction & operation; conventional-tube limits at microwave; bunching + BWO; slow backward wave structure + LNA (2+6)(2+8)(5)

## 6. RF Design Practices

- Filter — insertion loss method; Butterworth/Chebyshev prototype (5)(8)(4)(10)
- LPF/HPF/BPF via microstrip ( $\pi$ -section, T-type, series-arm, shunt-arm, double-section, triple-pad) +prototyping (3+5)(6+4)(10)(7+3)(2+8)(8+2)(5+3)(6+2)(3+8+3)
- Amplifier — stability + max gain (bilateral & unilateral), +Smith, +flowchart (5+5)(10)(12)(18)(7+8)(8+4)(4+4+4+4)(6+3+3)
  - S-params: 0.894∠-60.6; 0.55∠150; 0.656∠146.7; 0.72∠-116 (GaAs); 0.64∠-169; 0.55∠-160 (MESFET); 0.45∠163 (GaAs FET); 0.45∠-160; 0.78∠-113; 0.38∠-158 ( $Z_s=25$ ,  $Z_L=40$ ); 0.6∠-140; 0.55∠-150
  - flowchart (GaAsFET) +  $\Gamma_{in}/\Gamma_{out}$  trace; amplifier Vs oscillator (stability factor) (4)(5+5)(3+8)
- Oscillator & mixer theory (5)(8)

## 7. Microwave Antennas & Propagation

- RF/MW radiation hazards + fields; safety practices & standards (SARPs, IEEE, EMR) (8)(5)(6)(3+3)(5+5)
- SAR (definition, effect, non-ionizing); how radiation is hazardous; radiation zones of a microwave oven (6)(10)(4+4)(3+3)
- GSM BTS: propagation fields + radiation parameters; 900 Vs 1800 MHz fields; hazardous/safe zone @2600 MHz, 0.5 m antenna (4+4)(4+2+2)
- Reactive Vs near- Vs far-field; antenna as DUT — measurement tool + working principle; HERP (5+5)(8)(4)

## 8. RF/Microwave Measurements

- Power measurement — VSWR meter + low power / bolometry; VSWR > 10; measuring 7.5 mW; ASR power tool (2+8)(4+6)(2+6)(10)(8)
- Calorimeter — static / dry / circulating Vs flow; calorimeter wattmeter; static + dynamic circulating (5)(6)(7)(2+8)(6+2)(10)(6+3)
- Bolometer bridge (single-bridge limits, double-channel); bolometry principle; network analyzer (VNA); two-port model (8)(4)(10)(5)
- Microwave Vs low-frequency measurements; spectrum analyzer; device to map BTS radiation zones (3)(4)(5)(8)